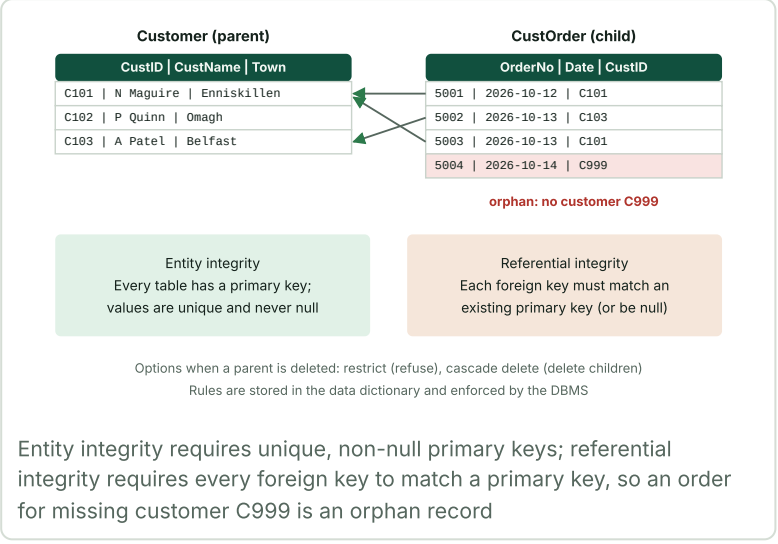
Data integrity: data dictionary, integrity rules, validation and indexing

IS04 FACT FILE

Specification: students should be able to

- Describe the purpose and contents of a data dictionary
- Explain entity integrity and referential integrity
- Describe how validation maintains data integrity
- Explain the purpose of indexing and its trade-offs

Key words: data integrity, data dictionary, metadata, entity integrity, referential integrity, cascade delete, orphan record, validation, verification, index, secondary key



Entity integrity requires unique, non-null primary keys; referential integrity requires every foreign key to match a primary key, so an order for missing customer C999 is an orphan record

Data dictionary

A data dictionary is a central store of metadata – data about the data. For every table it records the field names, data types and sizes, primary and foreign keys, validation rules, default values, indexes and relationships, and often the users who have access. Developers use it to understand the database and keep naming consistent; the DBMS uses it to enforce rules. Because the rules are held once, every application is forced to obey them.

Entity and referential integrity

Entity integrity requires every table to have a primary key whose values are unique and never null, so every record can be identified. Referential integrity requires every foreign key to match an existing primary key in the related table, preventing orphan records. The DBMS can enforce it by restricting deletions, by cascading deletions to child records or by cascading updates of key values. The right choice depends on the business rules.

Indexing

An index is a sorted data structure that maps values of one or more fields to the location of the records. It allows the DBMS to find records, sort and join tables much faster than by a full table scan, especially in large tables. However, indexes take extra storage and must be updated whenever a record is inserted, changed or deleted, so too many indexes slow down data entry. Good practice is to index fields that are frequently searched or used in joins.

T Key terms from the mark schemes

Data dictionary
A central store of metadata describing every table, field, data type, key, validation rule, index and relationship in a database.

Metadata
Data that describes other data, such as a field's name, type and size.

Entity integrity
The rule that every table has a primary key whose values are unique and never null.

Referential integrity
The rule that every foreign key value must match an existing primary key value in the related table, preventing orphan records.

Index
A sorted data structure of key values and pointers that lets the DBMS locate records quickly without scanning the whole table.

Cascade delete
A referential integrity option in which deleting a parent record automatically deletes its related child records.

Exam focus

Distinguish validation (checks data is reasonable and obeys rules) from verification (checks data matches the original source); neither guarantees accuracy. For integrity questions, name the rule and explain its effect in context: 'referential integrity would stop an Order being added for a CustID that does not exist in Customer'. For 'evaluate the use of indexes' give the benefit (faster search/join) and the cost (storage, slower updates) and reach a conclusion.

? Check yourself

1. State three items of metadata held in a data dictionary. [3]
2. Explain what is meant by entity integrity. [2]
3. Explain how referential integrity would prevent an error in a database containing Customer and Order tables. [3]

Answers: 1. Any three from: table names / field names (1) / data types (1) / field sizes / lengths (1) / primary keys (1) / foreign keys and relationships (1) / validation rules (1) / default values (1) / indexes (1) / users and access rights (1) / descriptions / formats (1). 2. Every table must have a primary key, and each primary key value must be unique (1) / and the primary key must never be null (empty), so every record can be identified unambiguously (1). 3. Referential integrity means every foreign key value must match an existing primary key value in the related table (1) / so the DBMS would reject an Order whose CustID does not exist in the Customer table, e.g. an order for C999 (1) / preventing orphan records; it would also stop (restrict) the deletion of a customer who still has orders, or cascade the change (1).